

MAHAVIR EDUCATION TRUST'S
Shah & Anchor Kutchhi Engineering College

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BE2-51

Date:

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SYSTEM SECURITY

ASSIGNMENT 3

Q1) Explain the following Web Browser Attacks:

- Account Harvesting
- Web Bugs
- Cross-site Request Forgery (CSRF)
- Session Hijacking

Ans

1. Account Harvesting

- Technique used to collect user credentials (username, passwords)
- Done using phishing emails or fake login pages
- Users are tricked into entering sensitive information
- Leads to unauthorized access and identity theft.

2. Web Bugs

- Small invisible images (1x1 pixel) or scripts in web pages / emails.
- Automatically send user data (IP address, browser details)
- Used for tracking user activity without consent
- Causes privacy invasion.

3. Cross-Site Request Forgery (CSRF)

- Tricks user's browser into performing unwanted actions.
- Occurs when user is already logged into a website.
- Malicious request is executed using stored cookies.
- Can result in unauthorized transactions or changes.

4. Session Hijacking

- Attacker steals session ID to take control of user session.
- Done via network sniffing or XSS attacks.
- No need for login credentials once session is captured.
- Leads to full account access.

- These attacks exploit browser security weaknesses.
- Can lead to data theft, privacy loss, and unauthorized access.
- Prevention includes HTTPS, secure cookies, and user awareness.

Q2) Describe IoT attacks and their countermeasures.

Ans IoT Attacks:

1. Botnet Attack (eg, Mirai)
 - IoT devices are infected and controlled remotely
 - Used to launch large-scale DDoS attacks.
2. Device Hijacking
 - Unauthorized access to IoT devices (cameras, smart devices)
 - Attacker can control or monitor the device.
3. Data Theft / Eavesdropping
 - Sensitive data is intercepted during transmission
 - Happens due to weak or no encryption
4. Malware Attacks
 - Malicious software infects IoT devices
 - Disrupts normal functioning or steals data
5. Denial of Service (DoS)
 - Flooding the device / network with traffic
 - Makes the system unavailable to users

Countermeasures:

1. Strong Authentication
 - Use strong passwords and multi-factor authentication.

2. Data encryption

- Encrypt data during transmission (Use HTTPS, TLS)

3. Regular Updates

- Keep firmware and software updated to fix vulnerabilities

4. Network Security

- Use firewalls, secure Wi-Fi, and network segmentation

5. Disable Unused Features

- Turn off unnecessary ports and services

Q3) Apply mobile security concepts to explain how smartphones can be protected from common threats

Ans

1. Strong authentication

- Use PIN, password, fingerprint, or face lock.
- Prevents unauthorized access to the device.

2. App Security

- Install apps from trusted sources (Play Store / App Store)
- Check app permission before installing
- Avoids malware and malicious apps.

3. Data encryption

- Encrypt stored data and use secure connections (HTTPS)
- Protects sensitive data from theft.

4. Regular Updates

- Keep OS and apps updated.
- Fixed security vulnerabilities and bugs.

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5. Network Security

- Avoid public Wi-Fi or use VPN
- Prevents data interception and hacking.

6. Anti-Virus / Security Apps

- Use mobile security software
- Detects and removes malware.

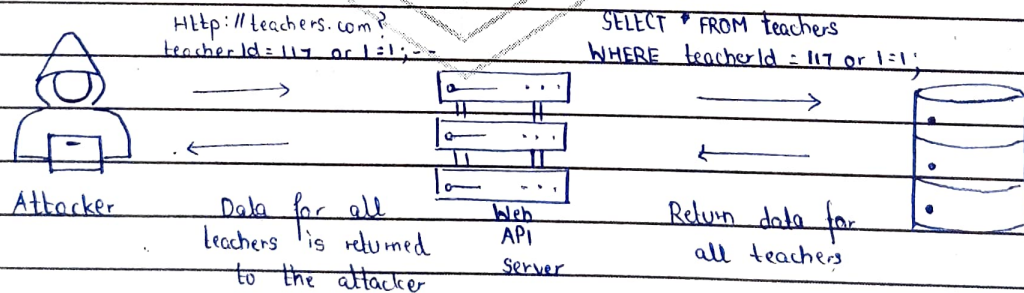
7. Backup & remote wipe

- Regularly backup data
- Use remote wipe in case of loss / theft.

Q4) Explain SQL injection attack and its prevention techniques

Ans SQL injection attack

- SQL injection is a web security attack where malicious SQL queries are inserted into input fields
- It targets the database of a web application
- Occurs when user input is not properly validated
- Attackers can view, modify, or delete database data
- Example: Entering ' OR '1'='1 to bypass login authentication



Prevention Techniques:

- Input Validation
 - Validate and sanitize all user inputs

- Use Prepared Statements (Parameterized Queries)
 - Prevents execution of malicious SQL code.
- Use Stored Procedures
 - Restrict direct database access
- Least Privilege Principle
 - Give minimum database access rights
- Error Handling
 - Do not display database errors to users
 - Prevents leakage of database structure.
- SQL injection can lead to data theft, modification, or complete system compromise
- Occurs mainly due to poor input validation and insecure coding
- Can be prevented using techniques like parameterized queries, validation and proper error handling
- Following secure coding practices is essential to protect web applications